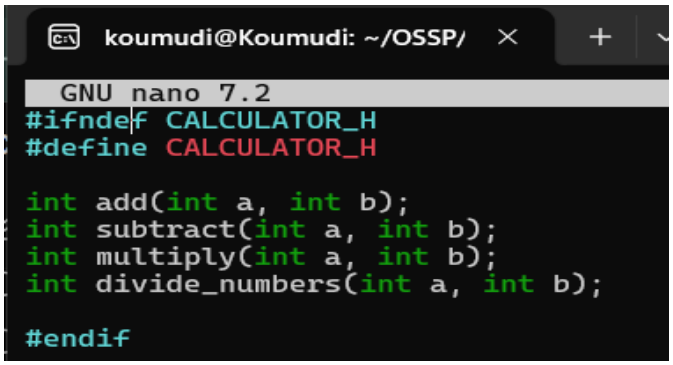
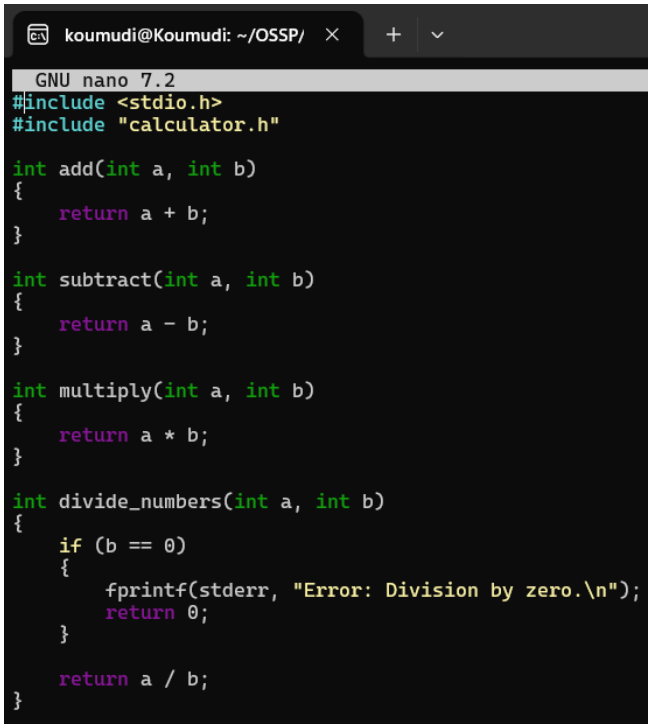
Skill OSSP

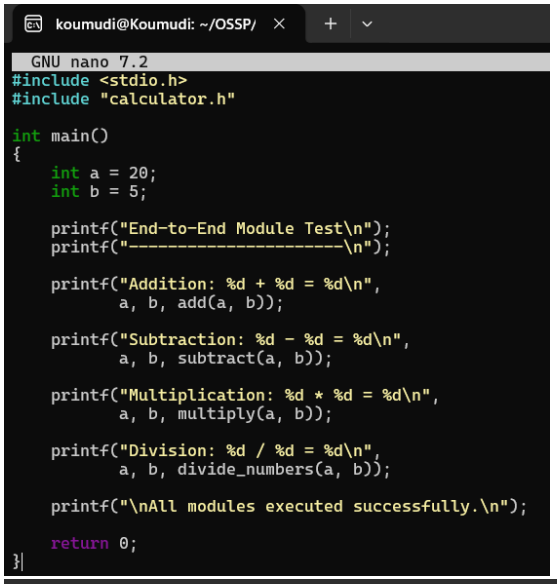
WEEK – 11 (S 21,22)

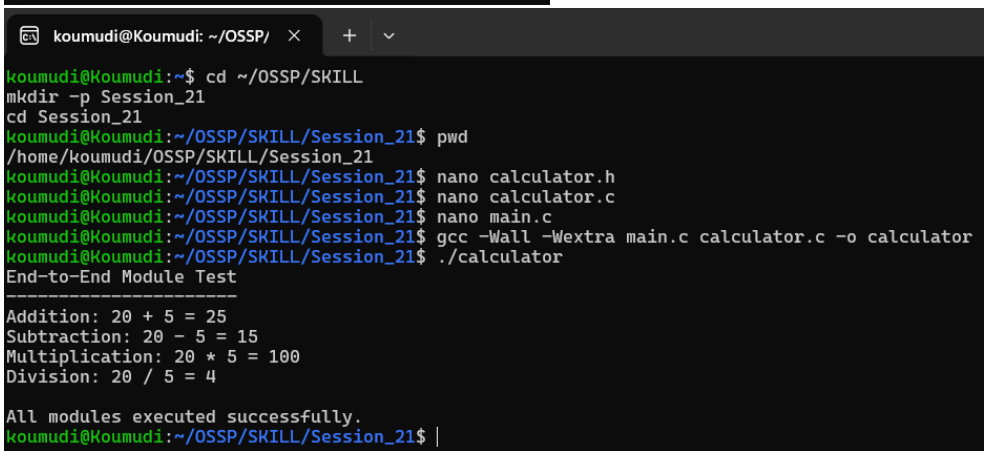
SESSION – 21:

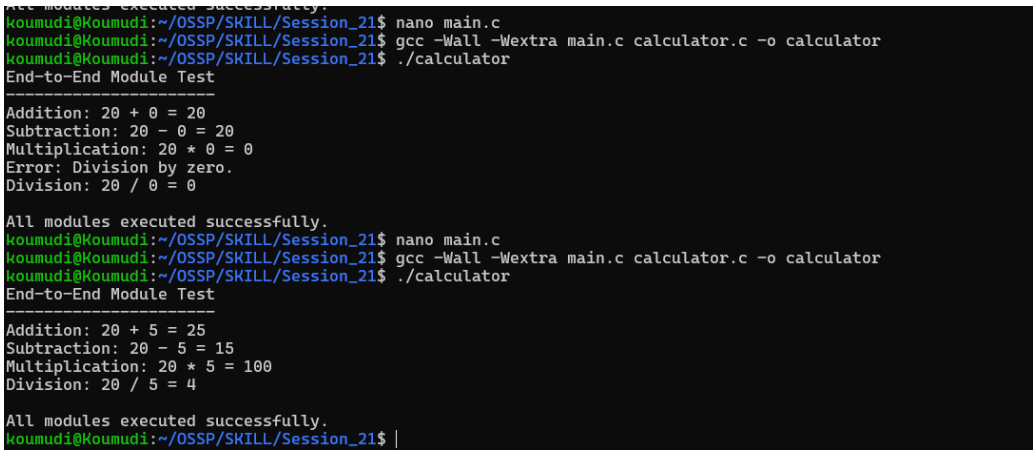
TASK1: To Connect Modules, Validate Interfaces, Test End-to-End Execution, Handle Errors, Improve Reliability, Refactor Components.











TASK2: TO Detect Runtime Errors, Generate Messages, Log Failures, Handle Invalid Syntax, Recover Gracefully, Test Error Scenarios.

koumudi@Koumudi: ~/OSSP/

GNU nano 7.2

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <errno.h>

void log_error(const char *message)
{
    FILE *log = fopen("error.log", "a");

    if (log == NULL)
    {
        perror("Unable to open error log");
        return;
    }

    fprintf(log, "ERROR: %s\n", message);
    fclose(log);
}

int main(int argc, char *argv[])
{
    printf("Runtime Error Handling Test\n");
    printf("---------------------------\n");

    if (argc < 2)
    {
        printf("Error: Missing input argument.\n");
        printf("Usage: %s <filename>\n", argv[0]);

        log_error("Missing input argument.");
        return 1;
    }

    FILE *file = fopen(argv[1], "r");

    if (file == NULL)
    {
        printf("Error: Unable to open file '%s'.\n",
               argv[1]);

        printf("Reason: %s\n", strerror(errno));
        printf("Error: Unable to open file '%s'.\n",
               argv[1]);

        printf("Reason: %s\n", strerror(errno));

        log_error("Failed to open requested file.");
        return 1;
    }

    printf("File '%s' opened successfully.\n", argv[1]);

    fclose(file);

    printf("Resources released successfully.\n");
    printf("Program recovered and exited normally.\n");

    return 0;
}
```

```
koumudi@Koumudi:~/OSSP/SKILL/Session_21$ nano error_handler.c
koumudi@Koumudi:~/OSSP/SKILL/Session_21$ gcc -Wall -Wextra error_handler.c -o error_handler
koumudi@Koumudi:~/OSSP/SKILL/Session_21$ ./error_handler
Runtime Error Handling Test
---------------------------
Error: Missing input argument.
Usage: ./error_handler <filename>
koumudi@Koumudi:~/OSSP/SKILL/Session_21$ ./error_handler missing.txt
Runtime Error Handling Test
---------------------------
Error: Unable to open file 'missing.txt'.
Reason: No such file or directory
koumudi@Koumudi:~/OSSP/SKILL/Session_21$ cat error.log
ERROR: Missing input argument.
ERROR: Failed to open requested file.
koumudi@Koumudi:~/OSSP/SKILL/Session_21$ echo "OSSP error handling test" > test.txt
koumudi@Koumudi:~/OSSP/SKILL/Session_21$ ./error_handler test.txt
Runtime Error Handling Test
---------------------------
File 'test.txt' opened successfully.
Resources released successfully.
Program recovered and exited normally.
koumudi@Koumudi:~/OSSP/SKILL/Session_21$
```

SESSION – 22:

TASK1: To Run Valgrind, Identify Leaks, Fix Memory Errors, Release Resources, Verify Cleanup, Document Findings.

koumudi@Koumudi: ~/OSSP/

```
  GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main()
{
    char *message;
    int *numbers;

    message = malloc(100 * sizeof(char));
    numbers = malloc(5 * sizeof(int));

    if (message == NULL || numbers == NULL)
    {
        printf("Memory allocation failed.\n");
        free(message);
        free(numbers);
        return 1;
    }

    strcpy(message, "OSSP Valgrind Memory Test");

    for (int i = 0; i < 5; i++)
    {
        numbers[i] = (i + 1) * 10;
    }

    printf("Message: %s\n", message);

    printf("Numbers: ");
    for (int i = 0; i < 5; i++)
    {
        printf("%d ", numbers[i]);
    }

    printf("\n");

    return 0;
}
```

```
koumudi@Koumudi:~/OSSP/SKILL/Session_22$ nano memory_test.c
koumudi@Koumudi:~/OSSP/SKILL/Session_22$ gcc -g -Wall -Wextra memory_test.c -o memory_test
koumudi@Koumudi:~/OSSP/SKILL/Session_22$ ./memory_test
Message: OSSP Valgrind Memory Test
Numbers: 10 20 30 40 50
koumudi@Koumudi:~/OSSP/SKILL/Session_22$ valgrind --version
valgrind-3.22.0
koumudi@Koumudi:~/OSSP/SKILL/Session_22$ valgrind --leak-check=full --track-origins=yes ./memory_test
==955== Memcheck, a memory error detector
==955== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==955== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==955== Command: ./memory_test
==955==
Message: OSSP Valgrind Memory Test
Numbers: 10 20 30 40 50
==955==
==955== HEAP SUMMARY:
==955==     in use at exit: 120 bytes in 2 blocks
==955==   total heap usage: 3 allocs, 1 frees, 1,144 bytes allocated
==955==
==955== 20 bytes in 1 blocks are definitely lost in loss record 1 of 2
==955==    at 0x4846828: malloc (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==955==    by 0x1091EC: main (memory_test.c:11)
==955==
==955== 100 bytes in 1 blocks are definitely lost in loss record 2 of 2
==955==    at 0x4846828: malloc (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==955==    by 0x1091DE: main (memory_test.c:10)
==955==
==955== LEAK SUMMARY:
==955==    definitely lost: 120 bytes in 2 blocks
==955==    indirectly lost: 0 bytes in 0 blocks
==955==      possibly lost: 0 bytes in 0 blocks
==955==    still reachable: 0 bytes in 0 blocks
==955==         suppressed: 0 bytes in 0 blocks
==955==
==955== For lists of detected and suppressed errors, rerun with: -s
==955== ERROR SUMMARY: 2 errors from 2 contexts (suppressed: 0 from 0)
koumudi@Koumudi:~/OSSP/SKILL/Session_22$
```

```
koumudi@Koumudi:~/OSSP/SKILL/Session_22$ nano memory_test.c
koumudi@Koumudi:~/OSSP/SKILL/Session_22$ gcc -g -Wall -Wextra memory_test.c -o memory_test
koumudi@Koumudi:~/OSSP/SKILL/Session_22$ ./memory_test
Message: OSSP Valgrind Memory Test
Numbers: 10 20 30 40 50
koumudi@Koumudi:~/OSSP/SKILL/Session_22$ valgrind --leak-check=full --track-origins=yes ./memory_test
==981== Memcheck, a memory error detector
==981== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==981== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==981== Command: ./memory_test
==981==
Message: OSSP Valgrind Memory Test
Numbers: 10 20 30 40 50
==981==
==981== HEAP SUMMARY:
==981==     in use at exit: 0 bytes in 0 blocks
==981==   total heap usage: 3 allocs, 3 frees, 1,144 bytes allocated
==981==
==981== All heap blocks were freed -- no leaks are possible
==981==
==981== For lists of detected and suppressed errors, rerun with: -s
==981== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
koumudi@Koumudi:~/OSSP/SKILL/Session_22$
```

TASK2: To Design Test Cases, Create Test Scripts, Automate Execution, Verify Outputs, Compare Results, Generate Reports.

koumudi@Koumudi: ~/OSSP/

```
 GNU nano 7.2
#include <stdio.h>

int add(int a, int b)
{
    return a + b;
}

int subtract(int a, int b)
{
    return a - b;
}

int multiply(int a, int b)
{
    return a * b;
}

int main()
{
    printf("ADD:%d\n", add(10, 5));
    printf("SUB:%d\n", subtract(10, 5));
    printf("MUL:%d\n", multiply(10, 5));

    return 0;
}
```

```
koumudi@Koumudi:~/OSSP/SKILL/Session_22$ nano calculator_test.c
koumudi@Koumudi:~/OSSP/SKILL/Session_22$ gcc -Wall -Wextra calculator_test.c -o calculator_test
koumudi@Koumudi:~/OSSP/SKILL/Session_22$ ./calculator_test
ADD:15
SUB:5
MUL:50
koumudi@Koumudi:~/OSSP/SKILL/Session_22$ nano test_script.sh
```

koumudi@Koumudi: ~/OSSP/

GNU nano 7.2

```
#!/bin/bash

echo "OSSP Automated Test Report"
echo "=========================="

PASS=0
FAIL=0

run_test()
{
    TEST_NAME=$1
    EXPECTED=$2

    ACTUAL=$(./calculator_test | grep "^$TEST_NAME")

    if [ "$ACTUAL" = "$EXPECTED" ]; then
        echo "PASS: $TEST_NAME"
        PASS=$((PASS + 1))
    else
        echo "FAIL: $TEST_NAME"
        echo "Expected: $EXPECTED"
        echo "Actual:   $ACTUAL"
        FAIL=$((FAIL + 1))
    fi
}

run_test "ADD:" "ADD:15"
run_test "SUB:" "SUB:5"
run_test "MUL:" "MUL:50"

echo
echo "=========================="
echo "Tests Passed: $PASS"
echo "Tests Failed: $FAIL"
echo "=========================="

if [ $FAIL -eq 0 ]; then
    echo "Overall Result: PASS"
else
    echo "Overall Result: FAIL"
fi
```

```
koumudi@Koumudi:~/OSSP/SKILL/Session_22$ nano test_script.sh
koumudi@Koumudi:~/OSSP/SKILL/Session_22$
koumudi@Koumudi:~/OSSP/SKILL/Session_22$ chmod +x test_script.sh
koumudi@Koumudi:~/OSSP/SKILL/Session_22$ ./test_script.sh
OSSP Automated Test Report
==========================
PASS: ADD:
PASS: SUB:
PASS: MUL:

==========================
Tests Passed: 3
Tests Failed: 0
==========================
Overall Result: PASS
koumudi@Koumudi:~/OSSP/SKILL/Session_22$ ./test_script.sh > test_report.txt
koumudi@Koumudi:~/OSSP/SKILL/Session_22$ cat test_report.txt
OSSP Automated Test Report
==========================
PASS: ADD:
PASS: SUB:
PASS: MUL:

==========================
Tests Passed: 3
Tests Failed: 0
==========================
Overall Result: PASS
koumudi@Koumudi:~/OSSP/SKILL/Session_22$
```